CSE400 Lecture 12 Scribe

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When you are writing the scribe, do reasoning step by step through definitions, assumptions, proofs, derivations, and worked example solutions exactly as presented in the provided context. Include this reasoning in the scribe so that the logical flow is clear for revision. Use only the provided lecture slides and relevant textbook excerpts. Don't mention any material, explanations or examples that are not present in the context.
Organize the scribe clearly in a formal academic style.

Below is a **formal lecture scribe suitable for exam preparation**, constructed **strictly from the provided lecture slides** (L11–L12: Transformation of Random Variables) without introducing outside material. Logical derivations and reasoning steps follow the sequence shown in the slides. Citations refer to the uploaded lecture PDF.

CSE400 – Fundamentals of Probability in Computing

Lecture Scribe: Transformation of Random Variables and Functions of Two Random Variables

Instructor

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SEAS – Ahmedabad University

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1. Introduction

This lecture studies **how probability distributions change when random variables are transformed**. The key goals are:

1. Learning techniques for **transformation of random variables**
2. Studying **functions of two random variables**
3. Understanding derived distributions through **illustrative examples such as** $Z = X + Y$

These topics are outlined in the lecture slides.

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2. Transformation of Random Variables

2.1 Problem Setup

Suppose we know the probability distribution of a random variable:

$$X \sim f_X(x)$$

Now define a **new random variable**

$$Y = g(X)$$

The goal is to determine the **distribution of** Y.

This requires deriving:

- the **CDF of** Y
- the **PDF of** Y

The lecture considers the case where $g(\cdot)$ **is monotonic**, which ensures that an **inverse function** $g^{-1}(\cdot)$ exists.

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3. CDF Method for Transformation

The transformation begins with the **definition of the cumulative distribution function (CDF)**.

Step 1: Write the CDF of Y

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

If g is monotonic increasing, we can apply the inverse transformation:

$$g(X) \leq y \Rightarrow X \leq g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

This step is shown conceptually in the transformation diagram of the lecture slides.

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4. Deriving the PDF

To obtain the **PDF**, differentiate the CDF with respect to y.

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substituting the previous expression:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Thus the **PDF transformation formula** becomes:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| \quad \text{where } x = g^{-1}(y)$$

This derivation corresponds to the steps illustrated in the lecture notes.

5. Worked Example: Transformation

Problem

Let

$$X \sim \text{Uniform}(-1, 1)$$

with

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Define

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Find $f_Y(y)$.

Step 1: Invert the Transformation

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Taking inverse sine:

$$X = \frac{2}{\pi}\sin^{-1}(Y)$$

Step 2: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3: Apply PDF Transformation

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute $f_X(x) = \frac{1}{2}$:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Final Result

$$f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1$$

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6. Function of Two Random Variables

The lecture next considers a transformation involving **two random variables**.

Define

$$Z = X + Y$$

The goal is to determine the **PDF of** Z.

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7. CDF Method for $Z = X + Y$

Start with the CDF definition.

$$F_Z(z) = P(Z \leq z)$$

Substitute $Z = X + Y$:

$$F_Z(z) = P(X + Y \leq z)$$

This probability corresponds to a **region in the** x, y **plane**, illustrated in the lecture diagram.

Thus,

$$F_Z(z) = \iint_{x+y \le z} f_{XY}(x, y)\, dx\, dy$$

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8. Using the Leibniz Rule

To obtain the PDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

The lecture applies the **Leibniz rule for differentiating integrals with variable limits**:

$$\frac{d}{dx} \int_{a(x)}^{b(x)} h(x, y) dy$$

This rule allows differentiation of the CDF integral.

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9. Derived Result

After applying the rule and simplifying:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x) dx$$

This formula expresses the **PDF of the sum of two random variables**.

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10. Special Case: Independent Variables

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Substitute into the previous result:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

This integral is called the **convolution integral**.

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11. Example: Sum of Normal Random Variables

Let

$$X \sim N(0,1)$$

$$Y \sim N(0,1)$$

and assume they are **independent**.

Using the convolution formula and simplifying the exponential terms shown in the slides:

$$f_Z(z) = \frac{1}{\sqrt{4\pi}} e^{-z^2/4}$$

Thus,

$$Z \sim N(0,2)$$

This shows that **the sum of two independent standard normal variables is a normal variable with variance 2**.

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12. Example: Sum of Exponential Variables

Let

$$X, Y \sim \text{Exponential}(\lambda)$$

with PDFs

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Using the convolution integral:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Simplifying:

$$f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z > 0$$

This result appears in the lecture derivation.

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13. Additional Transformations

The lecture also illustrates transformations such as:

$$Z = X - Y$$

$$Z = \frac{X}{Y}$$

and radial transformations like

$$R = \sqrt{X^2 + Y^2}$$

These examples highlight how **joint distributions are transformed to obtain new derived distributions**.

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14. Key Exam Takeaways

For exam preparation, the main results from the lecture are:

Single-variable transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of two random variables

General form:

$$f_Z(z) = \int f_{XY}(x, z - x) dx$$

Independent case (Convolution)

$$f_Z(z) = \int f_X(x) f_Y(z-x) dx$$

Important results

- $X, Y \sim N(0,1)$ independent
  $\rightarrow Z = X + Y \sim N(0,2)$
- Exponential RVs summed via convolution.

✔ These derivations and examples follow exactly the sequence and reasoning presented in the lecture slides.

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Below is a **formal exam-oriented lecture scribe** constructed **only from the uploaded lecture slides**. The reasoning, derivations, and examples follow the logical flow presented in the slides, with explanations written in a structured academic style for revision. All referenced material comes from the provided lecture document.

CSE400 – Fundamentals of Probability in Computing

Lecture Scribe: Transformation of Random Variables and Functions of Two Random Variables

Instructor

Dr. Dhaval Patel
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1. Lecture Overview

This lecture studies how the **probability distribution of a random variable changes when it undergoes a transformation**. The slides present three main components:

1. **Transformation of Random Variables**
   Learning techniques to determine the distribution of a transformed random variable.
2. **Function of Two Random Variables**
   Understanding transformations involving two random variables and their joint distributions.
3. **Illustrative Example**
   A detailed derivation for the case:

$$Z = X + Y$$

The objective is to derive the **probability density function (PDF)** of the transformed variable when the original distributions are known.

---

2. Transformation of a Random Variable

2.1 Problem Setup

Suppose a continuous random variable X has known PDF:

$$f_X(x)$$

and CDF:

$$F_X(x)$$

Now define a **new random variable**:

$$Y = g(X)$$

The task is to determine:

- $F_Y(y)$ — the CDF of Y
- $f_Y(y)$ — the PDF of Y

This scenario is illustrated graphically in the lecture slides, where Y is obtained by applying a function $g(\cdot)$ to X.

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3. Derivation of the Distribution Using the CDF Method

The derivation proceeds step by step.

Step 1: Write the CDF definition

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

If $g(\cdot)$ is **monotonic**, an inverse function g^{-1} exists.

Thus:

$$g(X) \leq y$$

is equivalent to

$$X \leq g^{-1}(y)$$

Hence,

$$F_Y(y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

This reasoning corresponds to the transformation diagram and derivation shown in the lecture slides.

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4. Deriving the PDF

To obtain the PDF, differentiate the CDF.

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute the previous expression:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Applying the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value to account for monotonic transformations:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| \quad \text{where } x = g^{-1}(y)$$

This gives the **general formula for PDF transformation**.

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5. Example: Transformation of a Uniform Random Variable

The lecture presents the following example.

Given

$$X \sim U(-1, 1)$$

Thus,

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Define

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Find $f_Y(y)$.

Step 1: Invert the transformation

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Taking inverse sine:

$$X = \frac{2}{\pi}\sin^{-1}(y)$$

Step 2: Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3: Apply the transformation formula

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

Substitute $f_X(x) = \frac{1}{2}$:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Final result

$$f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1$$

---

6. Function of Two Random Variables

The lecture next studies transformations involving **two random variables**.

Let:

$$Z = X + Y$$

The goal is to determine:

$$f_Z(z)$$

This is illustrated using a **geometric region in the x, y plane**, where the condition $x + y \leq z$ defines the integration region.

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7. Derivation Using the CDF Method

Start with the definition of the CDF.

$$F_Z(z) = P(Z \leq z)$$

Substitute $Z = X + Y$:

$$F_Z(z) = P(X + Y \leq z)$$

Express this probability as a **double integral over the joint PDF**:

$$F_Z(z) = \iint_{x+y\leq z} f_{XY}(x, y)\, dx\, dy$$

---

8. Differentiation Using Leibniz Rule

To obtain the PDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

The slides introduce the **Leibniz rule** for differentiation under the integral sign:

$$\frac{d}{dx} \int_{a(x)}^{b(x)} h(x, y) dy$$

This rule allows differentiation when the integration limits depend on the variable.

Applying the rule leads to:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx$$

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9. Independent Random Variables (Convolution)

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Substituting into the previous expression:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx$$

This integral is called the **convolution integral**.

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10. Example: Sum of Two Normal Random Variables

Let

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

and assume they are **independent**.

Applying the convolution formula and simplifying the exponential terms shown in the slides leads to:

$$f_Z(z) = \frac{1}{\sqrt{4\pi}} e^{-z^2/4}$$

Thus,

$$Z \sim N(0, 2)$$

Hence the **sum of two independent standard normal variables is normally distributed with variance 2**.

---

11. Example: Sum of Exponential Random Variables

Let

$$X, Y \sim \text{Exponential}(\lambda)$$

with

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$
$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Using convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Simplifying gives:

$$f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z > 0$$

---

12. Additional Transformations

The lecture briefly illustrates other transformations:

- Difference of variables

$$Z = X - Y$$

- Ratio of variables

$$Z = \frac{X}{Y}$$

- Radial transformation

$$R = \sqrt{X^2 + Y^2}$$

These examples demonstrate how **joint distributions can be transformed to derive new distributions**.

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13. Key Results for Examination

Single-variable transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of two random variables

$$f_Z(z) = \int f_{XY}(x, z - x)\, dx$$

Independent case (Convolution)

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Important examples

- $X, Y \sim N(0, 1)$ independent

$$Z = X + Y \sim N(0, 2)$$

- Sum of exponential variables derived using convolution.

✔ These results summarize the derivations and examples presented in the lecture slides.

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CSE400 – Fundamentals of Probability in Computing

Lecture Scribe: Transformation of Random Variables and Functions of Two Random Variables

Instructor

Dr. Dhaval Patel

Associate Professor, Computer Science and Engineering

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1. Lecture Overview

This lecture focuses on how the **distribution of a random variable changes when a transformation is applied**. The lecture contains three main components:

1. **Transformation of Random Variables**
   Techniques for deriving the distribution of a transformed variable.
2. **Function of Two Random Variables**
   Understanding transformations that involve two random variables.
3. **Illustrative Example**
   Detailed derivation for the transformation

$$Z = X + Y$$

The objective is to determine the **distribution of the transformed random variable when the original distributions are known**.

---

2. Transformation of a Random Variable

2.1 Problem Setup

Suppose we have a continuous random variable X whose distribution is known.

The slides indicate that both the **PDF and CDF of** X are known a priori.

$$f_X(x), \quad F_X(x)$$

Now define a **new random variable**

$$Y = g(X)$$

The goal is to determine:

- the **CDF of** Y, $F_Y(y)$
- the **PDF of** Y, $f_Y(y)$

The transformation process is illustrated graphically in the lecture slides (page 3), where a mapping $y = g(x)$ relates the values of X to Y.

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3. Derivation Using the CDF Method

The lecture derives the distribution of Y starting from the **definition of the CDF**.

Step 1: Definition of CDF

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

Step 2: Use the inverse transformation

If the function $g(\cdot)$ is **monotonically increasing**, the inverse function exists.

Thus the inequality

$$g(X) \leq y$$

can be rewritten as

$$X \leq g^{-1}(y)$$

Therefore

$$F_Y(y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

This step is illustrated in the diagram where the limits on the X-axis correspond to the inverse transformation of y.

---

4. Deriving the PDF

To obtain the PDF, differentiate the CDF.

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute the previous expression:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Applying the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Thus the transformation formula becomes

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| \quad \text{where } x = g^{-1}(y)$$

This formula is derived directly from the steps presented in the slides.

---

5. Worked Example: Transformation of a Uniform Random Variable

The lecture illustrates the method using the following example (page 5).

Given

$$X \sim U(-1, 1)$$

Thus

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Define

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Find $f_Y(y)$.

Step 1: Invert the transformation

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Applying the inverse sine function:

$$X = \frac{2}{\pi}\sin^{-1}(y)$$

Step 2: Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3: Apply the transformation formula

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

Substitute $f_X(x) = \frac{1}{2}$:

$$f_Y(y) = \frac{1}{2}\cdot\frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Final Result

$$f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1$$

---

6. Function of Two Random Variables

The lecture next studies transformations involving **two random variables**.

Define the new variable:

$$Z = X + Y$$

The objective is to determine the **PDF of** Z.

The diagram on page 7 shows the region

$$x + y \leq z$$

in the x-y plane.

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7. CDF of the Transformed Variable

Start from the CDF definition:

$$F_Z(z) = P(Z \leq z)$$

Substitute the transformation:

$$F_Z(z) = P(X + Y \leq z)$$

This probability corresponds to the shaded region shown in the slide diagram.

Thus

$$F_Z(z) = \iint_{x+y \leq z} f_{XY}(x, y)\, dx\, dy$$

---

8. Differentiation Using Leibniz Rule

To obtain the PDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

The lecture introduces the **Leibniz rule for differentiating integrals**:

$$\frac{d}{dx} \int_{a(x)}^{b(x)} h(x, y)\, dy$$

This rule allows differentiation when the integration limits depend on the variable.

Applying this rule yields

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx$$

---

9. Independent Random Variables (Convolution)

If X and Y are independent,

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Substituting into the previous expression gives

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx$$

This integral is called the **convolution integral**.

---

10. Example: Sum of Two Normal Random Variables

Let

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

with independence.

Applying the convolution formula and simplifying the expressions shown in the slides gives

$$f_Z(z) = \frac{1}{\sqrt{4\pi}} e^{-z^2/4}$$

Therefore

$$Z \sim N(0, 2)$$

This demonstrates that **the sum of two independent standard normal variables is normally distributed with variance 2**.

---

11. Example: Sum of Exponential Random Variables

Let

$$X, Y \sim \text{Exponential}(\lambda)$$

with PDFs

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$
$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Using convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Simplifying yields

$$f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z > 0$$

---

12. Additional Transformations

The slides also illustrate other transformations:

Difference

$$Z = X - Y$$

Ratio

$$Z = \frac{X}{Y}$$

Radial transformation

$$R = \sqrt{X^2 + Y^2}$$

These examples demonstrate how **joint random variables can be transformed into new random variables through functions of X and Y**.

---

13. Key Results for Exam Revision

Transformation of a single random variable

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Distribution of the sum of two random variables

$$f_Z(z) = \int f_{XY}(x, z - x) dx$$

Independent case (Convolution)

$$f_Z(z) = \int f_X(x) f_Y(z - x) dx$$

Important example results

$$X, Y \sim N(0, 1) \Rightarrow Z = X + Y \sim N(0, 2)$$

$$X, Y \sim \text{Exponential}(\lambda) \Rightarrow f_Z(z) = \lambda^2 z e^{-\lambda z}$$

✔ These derivations and results follow directly from the lecture slides and examples provided in the uploaded material.

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